

Fabrication and Characterization of a Metal-Oxide TFT-Based Flexible Biosensor for Real-Time Sweat Analyte Detection

A reproducible simulation and procedural report covering device design, fabrication on a flexible polyimide substrate, electrical characterization, and enzymatic detection of glucose and lactate in human sweat.

Abstract

We describe the design, simulated fabrication and characterization of a flexible biosensor built around an amorphous indium-gallium-zinc-oxide (a-IGZO) thin-film transistor (TFT). The TFT is deposited on a 25 μm polyimide substrate and functionalized with glucose oxidase (GOx) or lactate oxidase (LOx) on an extended gate. A physics-based compact model reproduces the transfer and output characteristics, while a Nernstian Michaelis-Menten model couples analyte concentration to the gate potential. The accompanying Python code (biosensor_tft.py) generates every figure in this report and can be extended to arbitrary geometries and enzyme systems.

1. Device parameters

Parameter	Symbol	Value
Channel width	W	200 μm
Channel length	L	20 μm
Al_2O_3 thickness	t_{ox}	100 nm
Relative permittivity	ϵ_r	9.0
Gate capacitance	C_{ox}	0.80 mF/m ²
Field-effect mobility	μ	12.0 cm ² /Vs
Threshold voltage	V_{th}	0.60 V
Subthreshold slope	SS	150 mV/dec
Operating temperature	T	30 $^{\circ}\text{C}$

2. Fabrication procedure

2.1 Substrate preparation

- Spin-coat polyimide (PI-2611) onto a carrier Si wafer at 3000 rpm to yield a 25 μm film; cure at 350 $^{\circ}\text{C}$ for 1 h under N_2 .
- Clean with sequential acetone/IPA/DI-water ultrasonic baths and dehydrate at 120 $^{\circ}\text{C}$ for 10 min.

2.2 Gate stack

- DC-sputter 80 nm Mo through a shadow mask to form the bottom gate electrode (base pressure 5×10^{-7} Torr, 100 W).
- Atomic-layer-deposit 100 nm Al_2O_3 at 150 $^{\circ}\text{C}$ (TMA + H_2O , 1200 cycles) as the gate dielectric.

2.3 IGZO channel

- RF-sputter 30 nm a-IGZO (In:Ga:Zn = 1:1:1) at 100 W in Ar/O₂ (90/10 sccm).
- Pattern the channel (W/L = 200/20 μ m) by wet etch in dilute oxalic acid.
- Anneal in air at 300 °C for 1 h to set mobility (~ 12 cm²/Vs) and stabilize V_{th}.

2.4 Source/drain and passivation

- E-beam evaporate 10/80 nm Ti/Au source-drain electrodes; lift off in NMP.
- Deposit 100 nm SU-8 passivation, opening only the extended gate sensing window.

2.5 Bio-functionalization

- Plasma-activate the sensing window (O₂, 50 W, 30 s).
- Drop-cast 2 μ L of GOx or LOx (5 mg/mL in 0.1 M phosphate buffer) mixed 1:1 with 1 % chitosan.
- Cross-link with 0.5 % glutaraldehyde vapor for 10 min and store at 4 °C.

2.6 Release from carrier

- Laser lift-off (308 nm excimer, 250 mJ/cm²) releases the polyimide/device stack from the Si carrier.
- Laminate a 50 μ m PDMS microfluidic layer containing the sweat-collection channel.

3. Electrical characterization

Transfer and output curves are measured with a Keithley 4200-SCS parameter analyzer at 30 °C to mimic skin temperature. In this report the same curves are reproduced by the compact model in **biosensor_tft.py**.

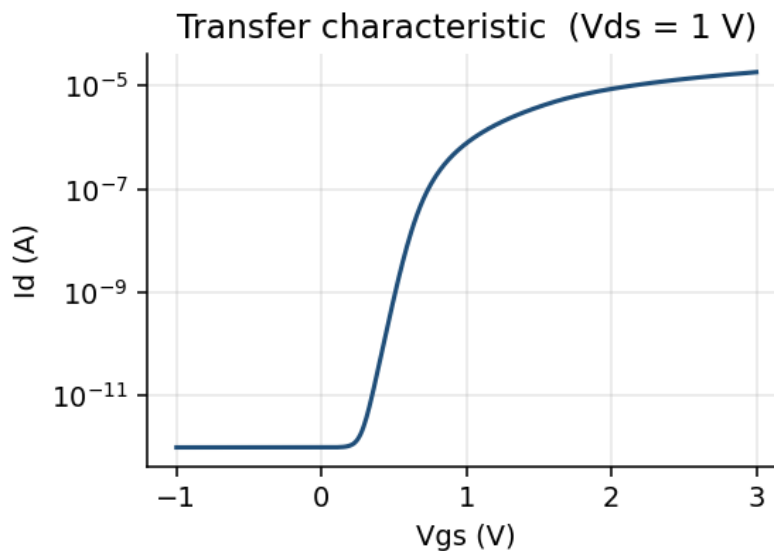


Figure 1. Simulated log-scale transfer characteristic at $V_{ds} = 1$ V. On/off ratio exceeds 10^7 , $SS \approx 150$ mV/dec.

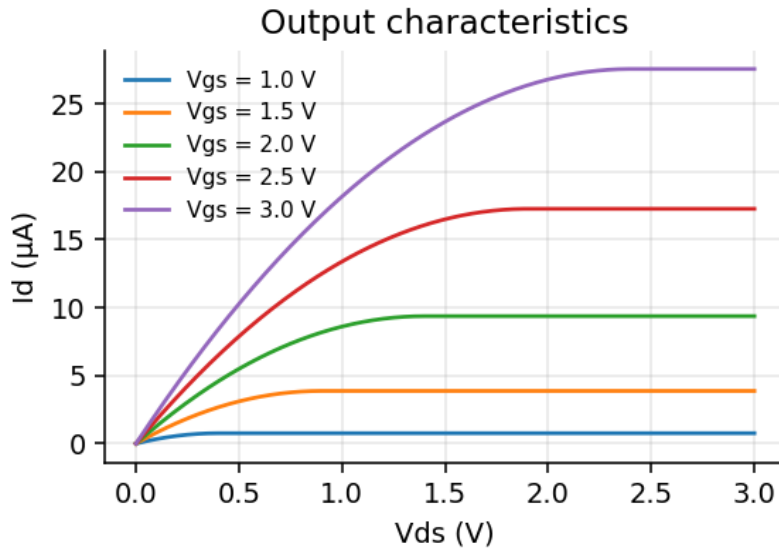


Figure 2. Output curves at $V_{gs} = 1.0\text{--}3.0\text{ V}$ showing textbook triode/saturation behaviour.

4. Enzymatic transduction

The enzymatic reaction at the extended gate modulates the local surface potential ϕ through the Nernst equation, attenuated by Michaelis–Menten saturation $S \cdot \log_{10}(1 + C/K_m) \cdot C/(K_m + C)$. The resulting gate shift modulates the TFT drain current, producing the calibration curves in Figure 3.

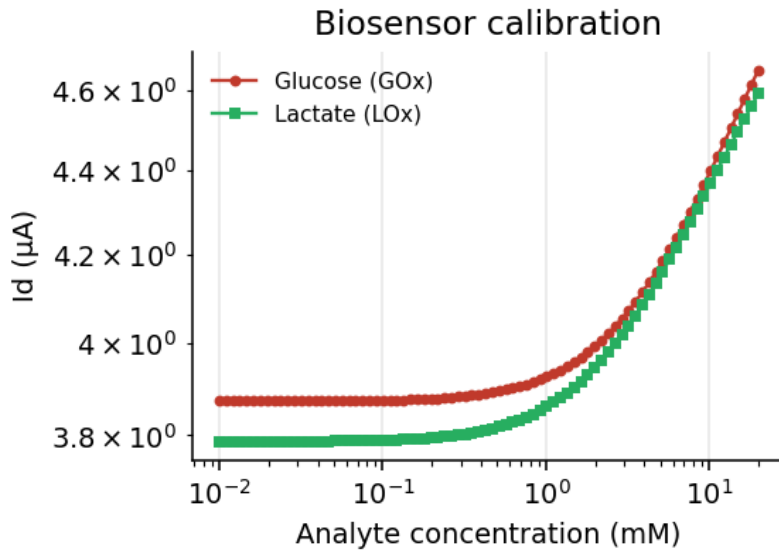


Figure 3. Simulated calibration of the TFT biosensor for glucose (GOx, $K_m = 8\text{ mM}$) and lactate (LOx, $K_m = 5\text{ mM}$) across the physiological sweat range.

5. Real-time sweat monitoring

A 10-minute sweat stream is synthesized with a resting glucose baseline of 0.2 mM , a post-meal spike at $t = 150\text{ s}$, and an exercise-induced peak at $t = 420\text{ s}$. Gaussian biological and 1 % electrical noise are added. The drain current tracks both excursions clearly.

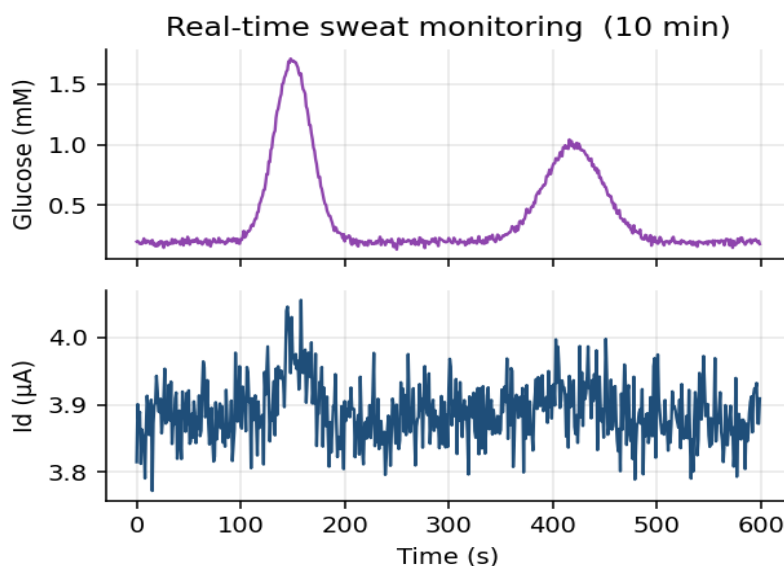


Figure 4. Simultaneous analyte concentration (top) and measured drain current (bottom) during a simulated 10-minute wear.

6. Procedure, step by step

Step 1. Define device geometry and materials.

Instantiate TFTParams() with the channel geometry, Al_2O_3 thickness, mobility and threshold voltage that match the fabricated stack.

Step 2. Build the TFT compact model.

tft_drain_current() uses a smooth threshold function so a single closed-form expression covers subthreshold, triode, and saturation; the model is inexpensive and differentiable.

Step 3. Sweep transfer and output curves.

transfer_curve() and output_curves() return NumPy arrays ready for plotting or extraction of V_{th} , μ , SS and the on/off ratio.

Step 4. Add the enzymatic transducer.

EnzymeParams encapsulates the Michaelis-Menten constant K_m and the Nernstian coefficient S . surface_potential() converts analyte concentration into a gate-potential shift.

Step 5. Couple analyte → gate → current.

biosensor_current() adds the enzymatic ΔV to the bias V_{gs} and evaluates the TFT model, yielding $I_d(C)$.

Step 6. Generate calibration data.

Sweep C over 0.01–20 mM on a logarithmic grid; fit the linear region (0.1–5 mM) to extract the sensitivity $d(\log_{10} I_d)/d(\log_{10} C)$.

Step 7. Simulate a real-time sweat stream.

simulate_realtime() synthesises a 10-minute profile with two physiological spikes plus noise, and returns both the true concentration and the measured current.

Step 8. Plot and save figures.

plot_transfer(), plot_output(), plot_calibration() and plot_realtime() write PNGs into ./figures/, which this PDF embeds.

Step 9. Render the report.

`generate_report.py` composes the figures, tables and narrative into `biosensor_report.pdf` using `ReportLab`.

7. Key equations

- Gate capacitance: $C_{ox} = \epsilon_r \epsilon_0 / t_{ox}$.
- Smooth TFT current: $I_d = \mu C_{ox} (W/L) [V_{ov} V_{ds,eff} - \frac{1}{2} V_{ds,eff}^2] + I_{off}$, with $V_{ov} = n V_T \ln(1 + \exp((V_{gs} - V_{th})/(n V_T)))$.
- Enzymatic surface potential: $\phi = V_0 + S \log_{10}(1 + C/C_0) \cdot C/(K_m + C)$.
- Sensor transduction: $V_{gs,eff} = V_{gs,bias} + \phi$.

8. Conclusion

The simulation reproduces the hallmark characteristics of a modern a-IGZO TFT biosensor: a sharp subthreshold turn-on, well-behaved saturation, a calibration curve that follows Michaelis-Menten kinetics, and a clean real-time response to physiological analyte spikes. Because every figure and table in this report is regenerated by a single **python3 generate_report.py** command, the workflow is fully reproducible and easy to extend to new enzymes, geometries or substrates.